PROBLEM SPECIFICATION

Delivery Forces

Universe Express Operations • Resource Optimization

Gry has finally been appointed executive head of logistics and courier operations at *Universe Express*. He has n couriers under his command, each having an individual delivery capacity f_i.

The delivery capacity of a three-person squad is defined as the **median** of the three members' capacities—that is, the middle value once their individual delivery capacities are sorted in non-decreasing order.

Help Gry partition all n couriers into k squads of three ($k = n / 3$) in order to maximize the overall delivery capacity of *Universe Express*. The overall delivery capacity is the sum of the capacities of all k squads.

INPUT

The first line contains a single integer n ($1 \le n \le 10^6$), representing the number of couriers in the company. It is guaranteed that n is a multiple of 3.

The second line contains n space-separated integers $f_1, f_2, \ldots, f_n$ ($1 \le f_i \le 10^6$), where f_i denotes the delivery capacity of the i-th courier.

OUTPUT

Print a single integer on a single line: the maximum total delivery capacity attainable by *Universe Express*.

SAMPLE INPUT 1	SAMPLE OUTPUT 1
3 1 2 3	2

Note on Sample 1: With $n = 3$ couriers, exactly one squad of three people can be formed: {1, 2, 3}. Sorting their capacities yields [1, 2, 3], whose median (middle element) is 2. Thus, the maximum total delivery capacity is 2.